INCH-POUND

MIL-S-80020D
31 January 1991
SUPERSEDING
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27 September 1989

MILITARY SPECIFICATION

SANDER, SINGLE AND DOUBLE DISK, WOODWORKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric motor driven, single and double disk, tilting table, floor mounted sanders with equipment and accessories specified herein. The requirements set forth in this specification and the completed ordering data (see paragraph 6.2), establishes the Government's minimum requirements for each procurement.

1.2 Classification. The sanders shall be of the following types and sizes. The types and sizes to be furnished shall be as specified (see 3.5 and 6.2).

Type 1 - Single Disk

Size 15 - 15-inch diameter disk
Size 20 - 20-inch diameter disk
Size 24 - 24-inch diameter disk
Size 30 - 30-inch diameter disk
Size 36 - 36-inch diameter disk
Size 37-1/2 - 37-1/2-inch diameter disk

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be ad dressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3220

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Type II - Double Disk

Size 20 - 20-inch diameter disk
Size 24 - 24-inch diameter disk
Size 30 - 30-inch diameter disk
Size 36 - 36-inch diameter disk
Size 37-1/2 - 37-1/2-inch diameter disk

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI O1.1a - Safety Requirements for Woodworking Machinery. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems, General Standards, for. (DOD adopted)

NEMA ICS 6 - Enclosures for Industrial Controls and Systems. (DOD adopted)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

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(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete sander shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The sanders shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The sanders shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the sanders shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

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3.2.3 Energy efficiency. The sanders and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical, electrical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the sanders, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI 01.1a. If a conflict arises between 29 CFR 1910 and ANSI 01.1a, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial disk sanders supplied to the commercial market shall be provided for the point of operation and all other parts of the sanders that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Environmental protection. During the manufacture, service, transportation, or storage of the machine and its components, the machine shall meet all applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection requirements exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

3.2.7 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.8 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective parts between end items of the same model. All replacement parts shall be interchangeable with their respective original parts, and fit/function with mating components.

3.3 Construction. The disk sanders shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the disk sanders to its maximum rating and capacity without permanent distortion or failure.

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3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the disk sanders and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4. Components. Sanders covered herein shall consist essentially of a frame and base, supporting a motor, table and such other components necessary to meet the requirements stated herein. Type 1 sanders shall have a single disk and a single tilting table to support hand-fed workpieces. Type II sanders shall have two disks and two tilting tables to support hand-fed workpieces.

3.4.1 Frame. The frame shall be of such mass and have sufficient strength and rigidity to support the working parts of the sander and hand-fed workpieces. The frame shall support and maintain alignment of the disk in relation to the workpiece support surface of the table. The frame shall have provisions on its base for attaching to the floor or foundation.

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3.4.2 Table. The table top surfaces and edges shall be finished and flat within +/-0.005 inch per foot. The table shall provide tilting characteristics as specified in table II. Table tilt shall be manually controlled and have a locking device, for positive retention, in the desired table tilt position. The table shall have vertical and horizontal adjustments conforming to the requirements of table 1. When the vertical adjustment range allows the table to be lowered below the bottom edge of the disk, to enable disk removal, the horizontal table adjustment may be omitted. The table shall have features as necessary for mounting miter and circular sanding attachments.

3.4.3 Disk. The disk shall be constructed of either carbon-steel plate or heat-treated cast aluminum having a tensile strength of not less than 25,000 PSI. The disk shall mount on an iron or steel hub in a secure and removable manner. The disk and hub shall be so constructed that concentricity and balance are maintained upon disk removal and reattachment. The disk face shall be finished for the adhesive attachment of an abrasive sanding disk.

3.4.4 Dust chute. The dust chute shall enclose the lower portion of the disk and be shaped for connection to an exhaust duct.

3.4.5 Drive. Unless otherwise specified (see 6.2), the sander shall be provided with either a direct or V-belt drive. Direct driven sanders shall have the hub mounted directly on the motor shaft. Belt driven sanders shall have means of maintaining belt alignment and tension.

3.4.6 Miter attachment. A miter attachment shall be fitted to each sander table. The miter attachment shall consist essentially of an adjustable fence attached to a metal alignment bar. The metal bar shall mate with a T-slot or groove, running the entire table length, parallel to the face of the disk. The fence shall have a swing adjustment of not less than 45 degrees on each side of the alignment bar. The swing adjustment range shall be clearly marked in degrees on or about the base of the fence.

3.4.7 Circular sanding attachment. When specified (see 6.2), a circular sanding attachment shall be provided to facilitate sanding of a circular workpiece with radii of one inch through the table dimension perpendicular to the sanding disk surface. The circular sanding attachment shall be constructed to mount on the tilting table and be infinitely adjustable to utilize the entire exposed surface of the sanding disk to produce both square and angular workpieces.

3.4.8 Electrical Controls. The sander shall be completely wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single source. Unless otherwise specified (see 6.2), the machine shall be designed for operation on 3-phase, 60-hertz (HZ), dual 230/460 volts. A foot operated start-stop switch shall be installed on the sander. When a reversible motor is specified (see 6.2), the controls shall provide for

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forward and reversing operation. The machine shall be initially wired for operation on the voltage specified by the procuring activity (see 6.2).

3.4.8.1 Motors. Motors shall be rated for continuous duty cycle and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1. Motor controls shall conform to the requirements of NEMA ICS-1 and NEMA ICS-6.

3.4.8.2 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the disk sander shall meet the requirements of table 1 for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. When specified (see 6.2), the physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Performance. The disk sander shall be capable of performing sanding operations on wood while utilizing the full capacity (down stroke side of the upper half of the disk) of the sanding disk. The sander shall be capable of performing the sanding operations with the table(s) set at each of the table tilt settings specified in table II. The sanded surfaces shall be flat and true with the table tilt settings, with no deviations in excess of 1/32 inch at every point.

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

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3.10 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant (if applicable)

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Workmanship. Workmanship of the disk sander and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality

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program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each disk sander prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. Each disk sander shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41^{0} to 104^{0} Fahrenheit and 20 to 95 percent relative humidity.

4.7.2 Operational test. Each disk sander shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

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4.7.3 Performance tests. Performance test shall be in accordance with 3.7 and table II. All performance tests shall be completed without performing any maintenance on the machine.

4.7.4 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.6. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.7.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The disk sanders covered by this specification are intended for use in woodworking shops where sander disk finishing and polishing operations are required.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

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g. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

h. Environmental protection requirements exceeding EPA/OSHA requirements (see 3.2.6).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

k. Drive, if different (see 3.4.5).

l. Circular sanding attachment, it required (see 3.4.7).

m. Electrical requirements, if different (see 3.4.8).

n. Reversible motor, if required (see 3.4.8).

o. Initial voltage, if different (see 3.4.8).

p. Motors, if different (see 3.4.8.1).

q. Worklight, if required (see 3.4.8.2).

r. Size and capacity, if different (see 3.5).

s. Specify physical size limitation, if required (see 3.5.1).

t. Specify and fully describe optional equipment, if required (see 3.8).

u. Specify applicable repair parts, if required (see 3.9).

v. Lubrication plate, if required (see 3.10.1).

w. Information to be provided on lubrication plate, if different (see 3.10.1).

x. Nameplate, if different (see 3.10.2).

y. Inspection conditions, if different (see 4.3).

z. First article inspection, if different (see 4.4).

aa. Quality conformance inspection, if different (see 4.5).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
6.3.1	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language
6.3.1	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined

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by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

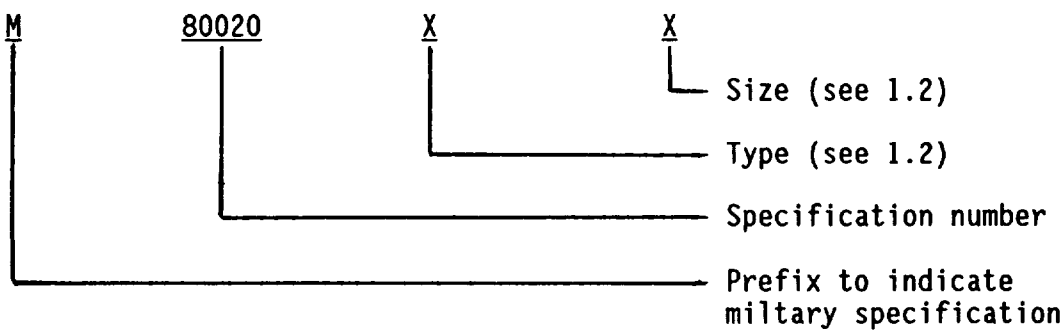
6.8 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



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6.13 Subject term (keyword) listing.

Electric
Floor-mounted
Machine
Miter
Polishing
Sanding, wood
Shop, wood

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TABLE 1. Characteristics, type and size.
Type 1

Characteristics	Size 15	Size 20	Size 24	Size 30	Size 36	Size 37-1/2
Disk:						
Diameter, inches	15	20	24	30	36	37-1/2
Thickness, inches	3/8	1/2	1/2	5/8	3/4	5/8
Speed range, RPM (no load)	1700 to 1800	1200 to 1800	1200 to 1300	800 to 1200	800 to 1200	800 to 900
Table:						
size, table top inches	21x9	25x12	30x12	40x15	46x20	20x46
Vertical adj. inches	1	4	6	8	8	6
Horizontal adj. to and from disk, inches	2	2-3/4	2	3	3	4
Motor horsepower Direct drive	2	2	2	5	7-1/2	7-1/2

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TABLE 1. Characteristics, type and size cont.
Type II

Characteristics	Size 20	Size 24	Size 30	Size 36	Size 37-1/2
Disk:					
Diameter, inches	20	24	30	36	37-1/2
Thickness, inches	1/2	9/16	5/8	3/4	5/8
Speed range, RPM (no load)	1200 to 1800	1100 to 1250	1000 to 1200	800 to 900	800 to 900
Table:					
size, table top inches	11x25	12x31	14x38	20x48	20x46
Vertical adj. inches	4	8	8	6	6
Horizontal adj. to and from disk, inches	2-3/4	3	3	4	4
Motor horsepower					
Direct drive	2	3	5	7-1/2	7-1/2
Belt drive	3	5	5	7-1/2	___

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TABLE II. Performance Requirements.

Work Table Tilt Setting	Resulting Sanding Angle (Degrees)
0 Degrees (no tilt)	90 +/- 1/2
10 Degrees toward the disk (up)	80 +/- 1/2
22 Degrees away from the disk (down)	112 +/- 1/2
45 Degrees away from the disk (down)	135 +/- 1/2

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activities:
Navy - YD
Air Force - 84
DLA - GS

User activities:
Army - AV, CE
Navy - MC

Preparing activity:
DLA - IP

Project (3220-0161)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-80020C	910131

3. DOCUMENT TITLE
SANDER, SINGLE AND DOUBLE DISK, WOODWORKING

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSM	(1) Commercial 901-775-4797	(2) AUTOVON AV 683-4797
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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